

Machine Learning Is Learning Feasible?

Prof. Shuyuan Yang, Zhixi Feng

E-mail: syyang@xidian.edu.cn

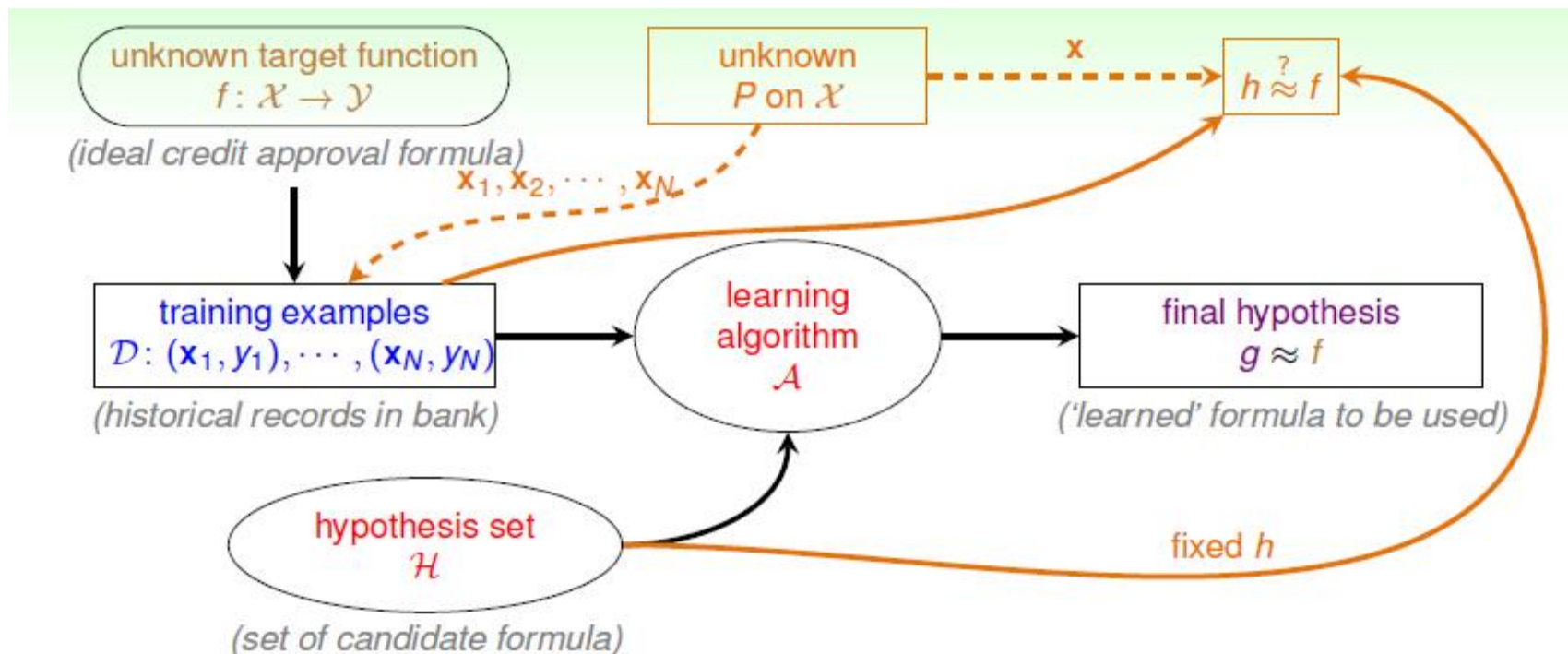
zxfeng@xidian.edu.cn



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- Is learning feasible?

Learning?



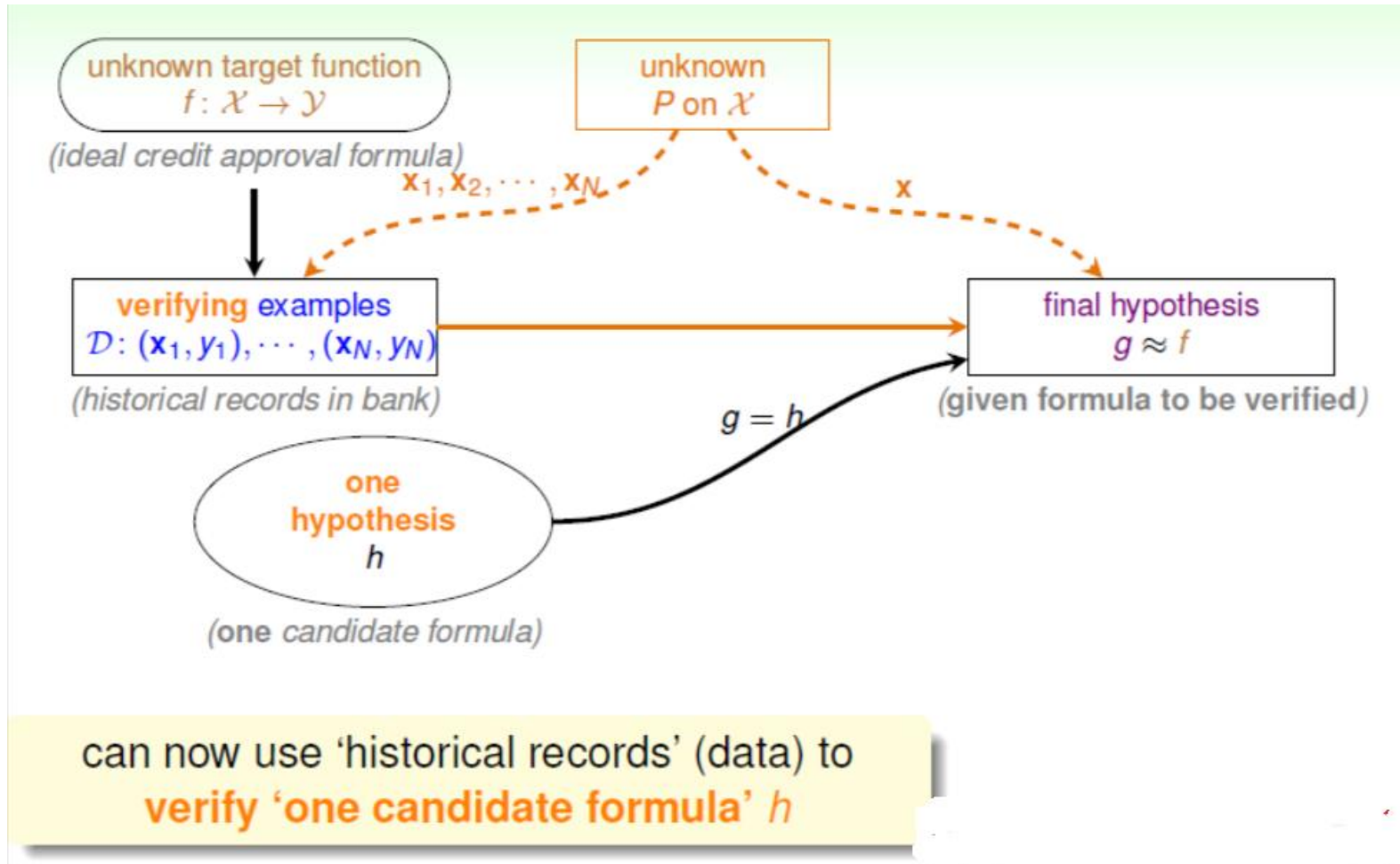
for any fixed h , can probably infer

$$\text{unknown } E_{\text{out}}(h) = \mathbb{E}_{\mathbf{x} \sim P} [\mathbb{I}[h(\mathbf{x}) \neq f(\mathbf{x})]]$$

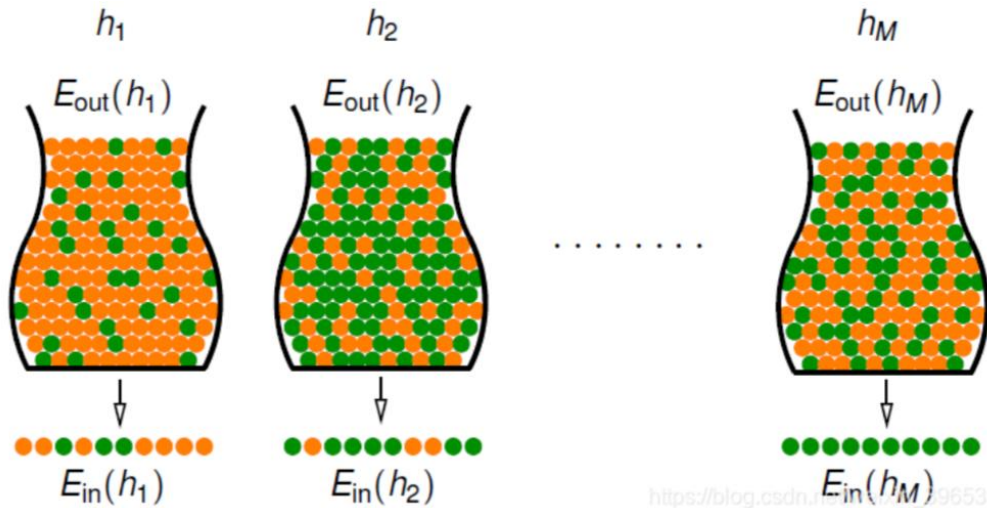
$$\text{by known } E_{\text{in}}(h) = \frac{1}{N} \sum_{n=1}^N [\mathbb{I}[h(\mathbf{x}_n) \neq y_n]].$$

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Learning?



Learning?



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	$\mathcal{D}_1$	$\mathcal{D}_2$	...	$\mathcal{D}_{1126}$	...	$\mathcal{D}_{5678}$	...	Hoeffding
h	BAD					BAD		$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D} \text{ for } h] \leq \dots$

Hoeffding: small

$$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D}] = \sum_{\text{all possible } \mathcal{D}} \mathbb{P}(\mathcal{D}) \cdot [\text{BAD } \mathcal{D}]$$

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Learning?

BAD data for many h

$\iff$ **no 'freedom of choice'** by $\mathcal{A}$

$\iff$ **there exists some h such that** $E_{\text{out}}(h)$ and $E_{\text{in}}(h)$ far away

	$\mathcal{D}_1$	$\mathcal{D}_2$	...	$\mathcal{D}_{1126}$	...	$\mathcal{D}_{5678}$	Hoeffding
h_1	BAD					BAD	$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D} \text{ for } h_1] \leq \dots$
h_2		BAD					$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D} \text{ for } h_2] \leq \dots$
h_3	BAD	BAD				BAD	$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D} \text{ for } h_3] \leq \dots$
...							
h_M	BAD					BAD	$\mathbb{P}_{\mathcal{D}} [\text{BAD } \mathcal{D} \text{ for } h_M] \leq \dots$
all	BAD	BAD				BAD	?

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Learning?

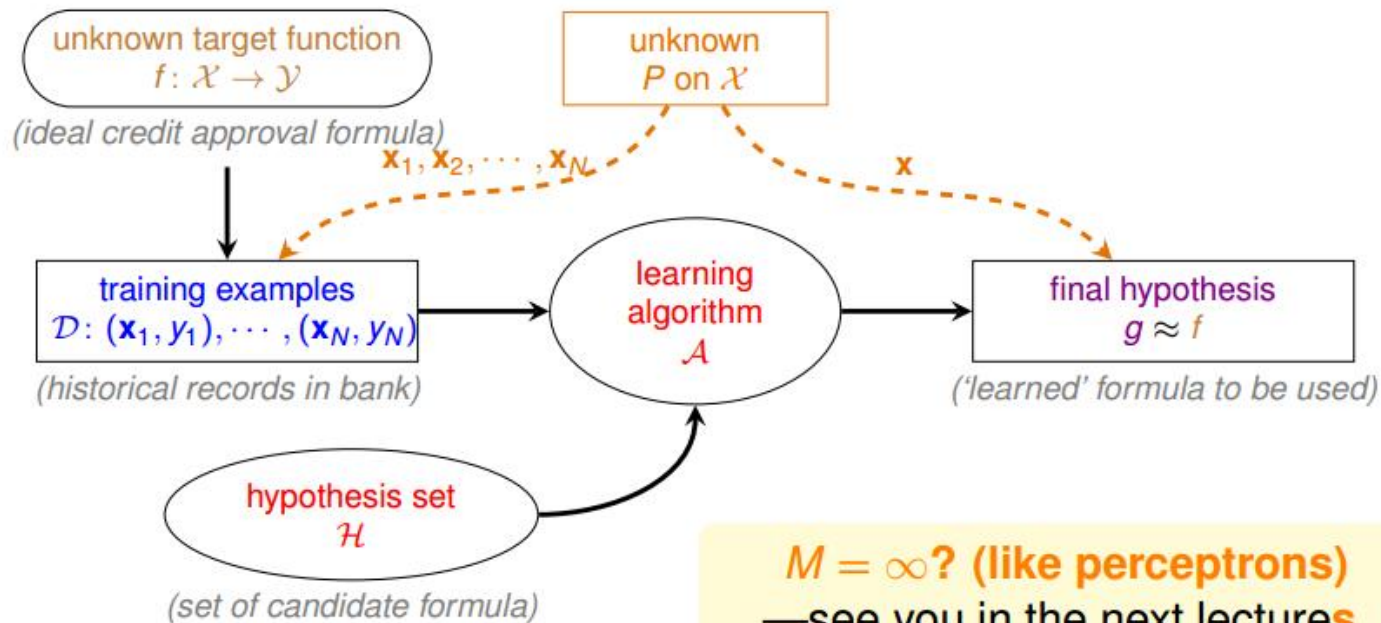
$$\begin{aligned} & \mathbb{P}_{\mathcal{D}}[\text{BAD } \mathcal{D}] \\ = & \mathbb{P}_{\mathcal{D}}[\text{BAD } \mathcal{D} \text{ for } h_1 \text{ or BAD } \mathcal{D} \text{ for } h_2 \text{ or } \dots \text{ or BAD } \mathcal{D} \text{ for } h_M] \\ \leq & \mathbb{P}_{\mathcal{D}}[\text{BAD } \mathcal{D} \text{ for } h_1] + \mathbb{P}_{\mathcal{D}}[\text{BAD } \mathcal{D} \text{ for } h_2] + \dots + \mathbb{P}_{\mathcal{D}}[\text{BAD } \mathcal{D} \text{ for } h_M] \\ & \text{(union bound)} \\ \leq & 2 \exp(-2\epsilon^2 N) + 2 \exp(-2\epsilon^2 N) + \dots + 2 \exp(-2\epsilon^2 N) \\ = & 2M \exp(-2\epsilon^2 N) \end{aligned}$$

- finite-bin version of Hoeffding, valid for all M , N and ϵ
- does not depend on any $E_{\text{out}}(h_m)$, **no need to 'know' $E_{\text{out}}(h_m)$** — f and P can stay unknown
- ' $E_{\text{in}}(g) = E_{\text{out}}(g)$ ' is **PAC**, **regardless of $\mathcal{A}$**

'most reasonable' $\mathcal{A}$ (like PLA/pocket):
pick the h_m with **lowest $E_{\text{in}}(h_m)$** as g

Learning?

if $|\mathcal{H}| = M$ finite, N large enough,
for whatever g picked by $\mathcal{A}$, $E_{\text{out}}(g) \approx E_{\text{in}}(g)$
if $\mathcal{A}$ finds one g with $E_{\text{in}}(g) \approx 0$,
PAC guarantee for $E_{\text{out}}(g) \approx 0 \implies$ **learning possible :-)**



Two Central Questions

learning split to two central questions:

- 1 can we make sure that $E_{\text{out}}(g)$ is close enough to $E_{\text{in}}(g)$?
- 2 can we make $E_{\text{in}}(g)$ small enough?

Trade-off on M

- 1 can we make sure that $E_{\text{out}}(g)$ is close enough to $E_{\text{in}}(g)$?
- 2 can we make $E_{\text{in}}(g)$ small enough?

small M

- 1 Yes!,
 $\mathbb{P}[\mathbf{BAD}] \leq 2 \cdot M \cdot \exp(\dots)$
- 2 No!, too few choices

large M

- 1 No!,
 $\mathbb{P}[\mathbf{BAD}] \leq 2 \cdot M \cdot \exp(\dots)$
- 2 Yes!, many choices

using the right M (or $\mathcal{H}$) is important
 $M = \infty$ **doomed?**

Preview

Known

$$\mathbb{P} [|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon] \leq 2 \cdot M \cdot \exp(-2\epsilon^2 N)$$

Todo

- establish **a finite quantity** that replaces M

$$\mathbb{P} [|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon] \stackrel{?}{\leq} 2 \cdot m_{\mathcal{H}} \cdot \exp(-2\epsilon^2 N)$$

- justify the feasibility of learning for infinite M
- study $m_{\mathcal{H}}$ to understand its trade-off for ‘right’ $\mathcal{H}$, just like M

mysterious PLA to be fully resolved
after 3 more lectures :-)

Where Did ***M*** Come From?

$$\mathbb{P} [|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon] \leq 2 \cdot \mathbf{M} \cdot \exp(-2\epsilon^2 N)$$

- **BAD events** $\mathcal{B}_m$: $|E_{\text{in}}(h_m) - E_{\text{out}}(h_m)| > \epsilon$
- to give $\mathcal{A}$ freedom of choice: bound $\mathbb{P}[\mathcal{B}_1 \text{ or } \mathcal{B}_2 \text{ or } \dots \mathcal{B}_M]$
- worst case: all $\mathcal{B}_m$ non-overlapping

$$\mathbb{P}[\mathcal{B}_1 \text{ or } \mathcal{B}_2 \text{ or } \dots \mathcal{B}_M] \underbrace{\leq}_{\text{union bound}} \mathbb{P}[\mathcal{B}_1] + \mathbb{P}[\mathcal{B}_2] + \dots + \mathbb{P}[\mathcal{B}_M]$$

where did **uniform bound fail**
to consider for $M = \infty$?

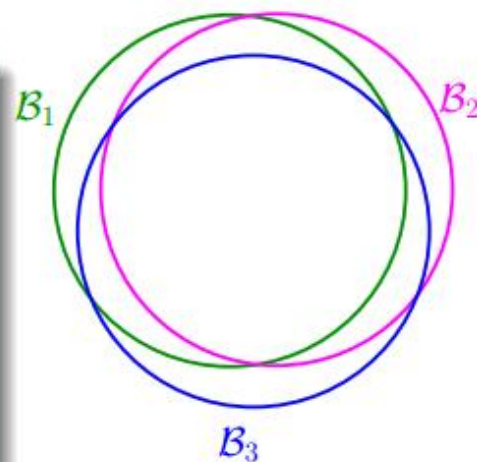
Where Did ***M*** Come From?

union bound $\mathbb{P}[\mathcal{B}_1] + \mathbb{P}[\mathcal{B}_2] + \dots + \mathbb{P}[\mathcal{B}_M]$

- **BAD events** $\mathcal{B}_m$: $|E_{\text{in}}(h_m) - E_{\text{out}}(h_m)| > \epsilon$

overlapping for similar hypotheses $h_1 \approx h_2$

- why? ① $E_{\text{out}}(h_1) \approx E_{\text{out}}(h_2)$
② for most $\mathcal{D}$, $E_{\text{in}}(h_1) = E_{\text{in}}(h_2)$
- union bound **over-estimating**

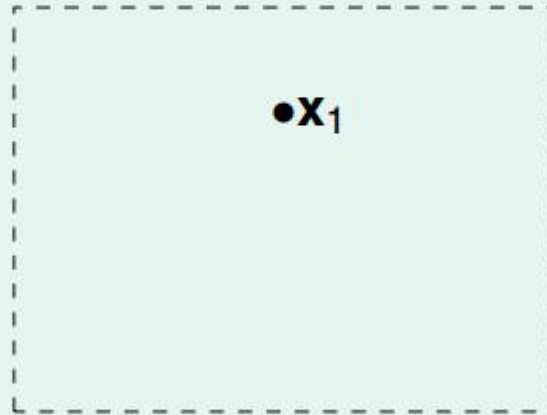


to account for overlap,
can we group similar hypotheses by **kind**?

How Many Lines Are There? (1/2)

$$\mathcal{H} = \left\{ \text{all lines in } \mathbb{R}^2 \right\}$$

- how many lines? ∞
- how many **kinds of** lines if viewed from one input vector $\mathbf{x}_1$?

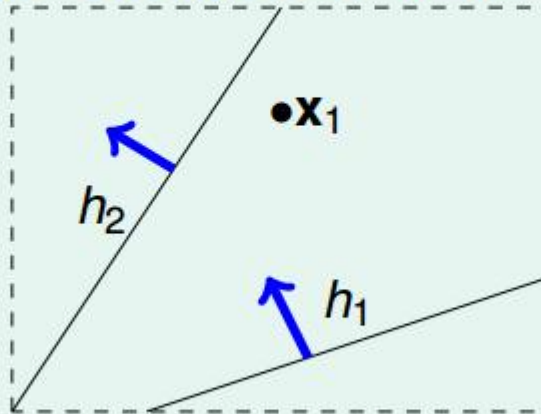


2 kinds: $h_1\text{-like}(\mathbf{x}_1) = \circ$ or $h_2\text{-like}(\mathbf{x}_1) = \times$

How Many Lines Are There? (1/2)

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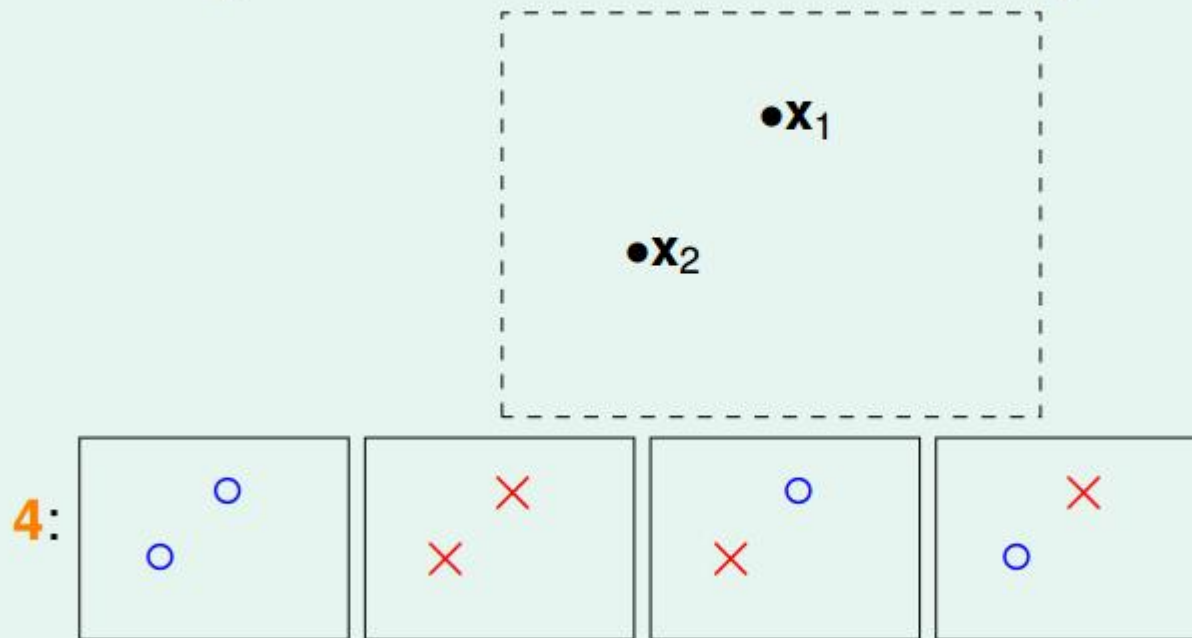


2 kinds: h_1 -like($\mathbf{x}_1$) = $\circ$ or h_2 -like($\mathbf{x}_1$) = $\times$

How Many Lines Are There? (2/2)

$$\mathcal{H} = \{ \text{all lines in } \mathbb{R}^2 \}$$

- how many **kinds of** lines if viewed from two inputs $\mathbf{x}_1, \mathbf{x}_2$?

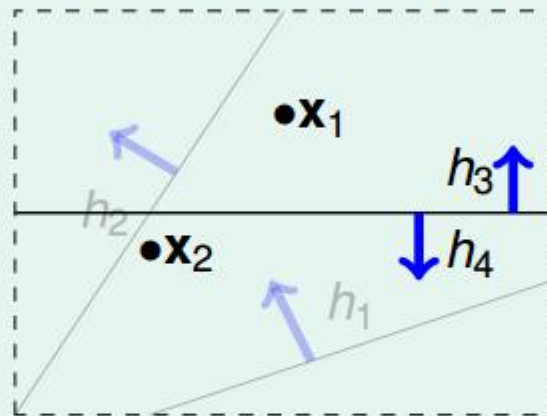


one input: 2; two inputs: 4; **three inputs?**

How Many Lines Are There? (2/2)

$$\mathcal{H} = \left\{ \text{all lines in } \mathbb{R}^2 \right\}$$

- how many **kinds of** lines if viewed from two inputs $\mathbf{x}_1, \mathbf{x}_2$?



4:

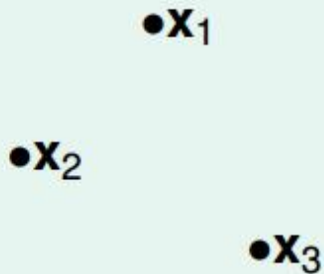


one input: 2; two inputs: 4; **three inputs?**

How Many Kinds of Lines for Three Inputs? (1/2)

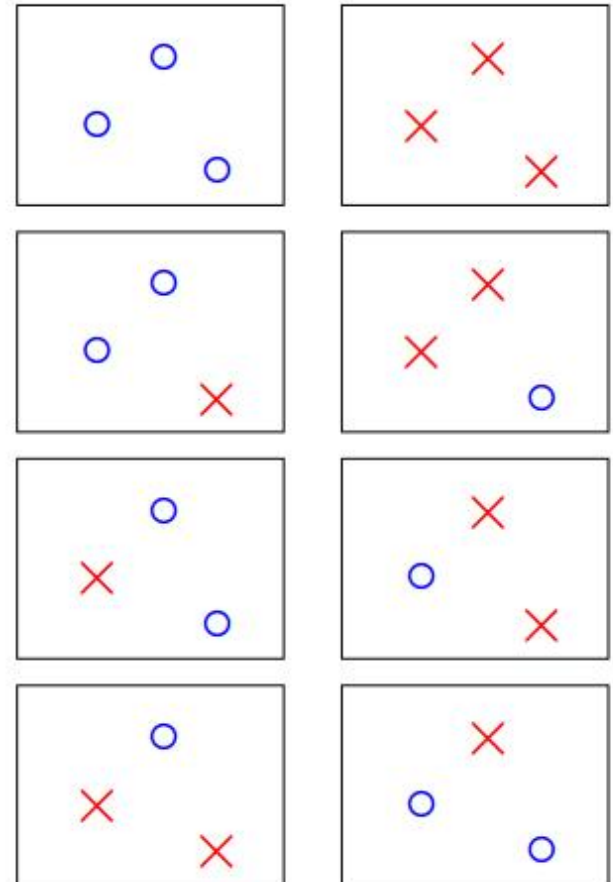
$$\mathcal{H} = \{\text{all lines in } \mathbb{R}^2\}$$

for three inputs $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$



always **8** for three inputs?

8:

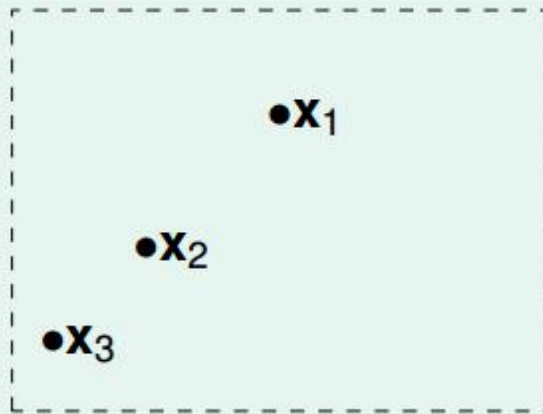


How Many Kinds of Lines for Three Inputs? (2/2)

$$\mathcal{H} = \{ \text{all lines in } \mathbb{R}^2 \}$$

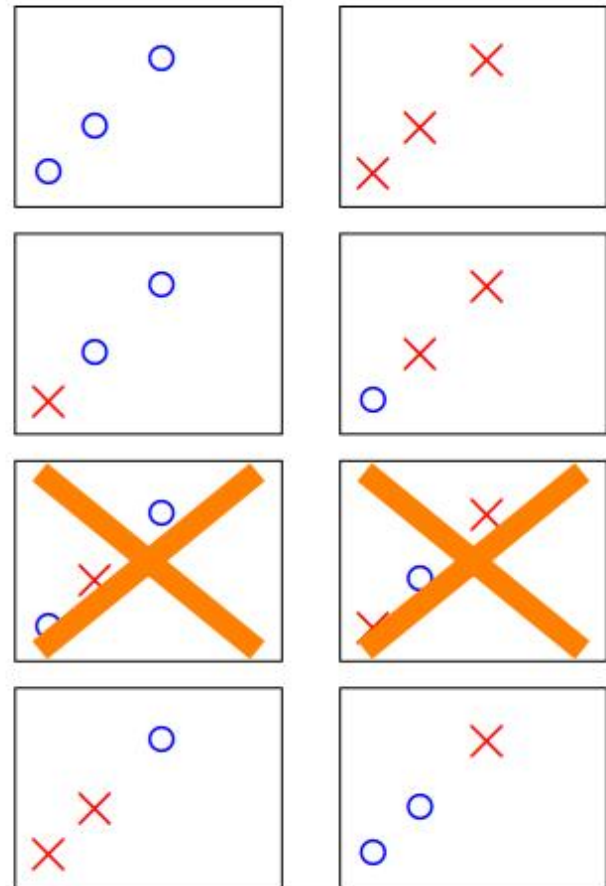
for **another** three inputs

$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$



'fewer than 8' when degenerate
(e.g. collinear or same inputs)

6:



Effective Number of Lines

maximum kinds of lines with respect to N inputs $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$
 $\iff$ **effective number of lines**

- must be $\leq 2^N$ (why?)
- finite 'grouping' of infinitely-many lines $\in \mathcal{H}$
- wish:

$$\begin{aligned} & \mathbb{P} [|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon] \\ & \leq 2 \cdot \text{effective}(N) \cdot \exp(-2\epsilon^2 N) \end{aligned}$$

lines in 2D

N	effective(N)
1	2
2	4
3	8
4	$14 < 2^N$

if ① effective(N) can replace M and

② effective(N) $\ll 2^N$

learning possible with infinite lines :-)

Fun Time

Consider 4 hypotheses.

$$h_1(\mathbf{x}) = \text{sign}(x_1), \quad h_2(\mathbf{x}) = \text{sign}(x_2),$$

$$h_3(\mathbf{x}) = \text{sign}(-x_1), \quad h_4(\mathbf{x}) = \text{sign}(-x_2).$$

For any N and ϵ , which of the following statement is not true?

- ① the **BAD** data of h_1 and the **BAD** data of h_2 are exactly the same
- ② the **BAD** data of h_1 and the **BAD** data of h_3 are exactly the same
- ③ $\mathbb{P}_{\mathcal{D}}[\text{BAD for some } h_k] \leq 8 \exp(-2\epsilon^2 N)$
- ④ $\mathbb{P}_{\mathcal{D}}[\text{BAD for some } h_k] \leq 4 \exp(-2\epsilon^2 N)$

Fun Time

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- ④ $\mathbb{P}_{\mathcal{D}}[\text{BAD for some } h_k] \leq 4 \exp(-2\epsilon^2 N)$

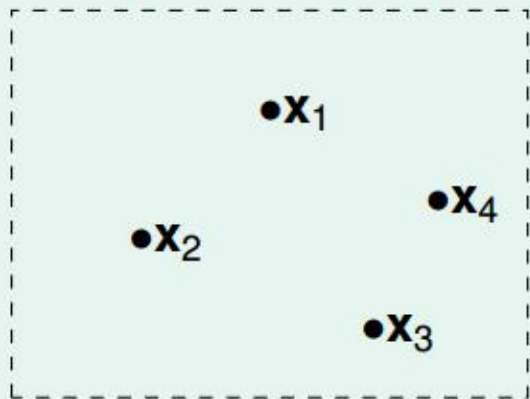
Reference Answer: ①

The important thing is to note that ② is true, which implies that ④ is true if you revisit the union bound. Similar ideas will be used to conquer the $M = \infty$ case.

How Many Kinds of Lines for Four Inputs?

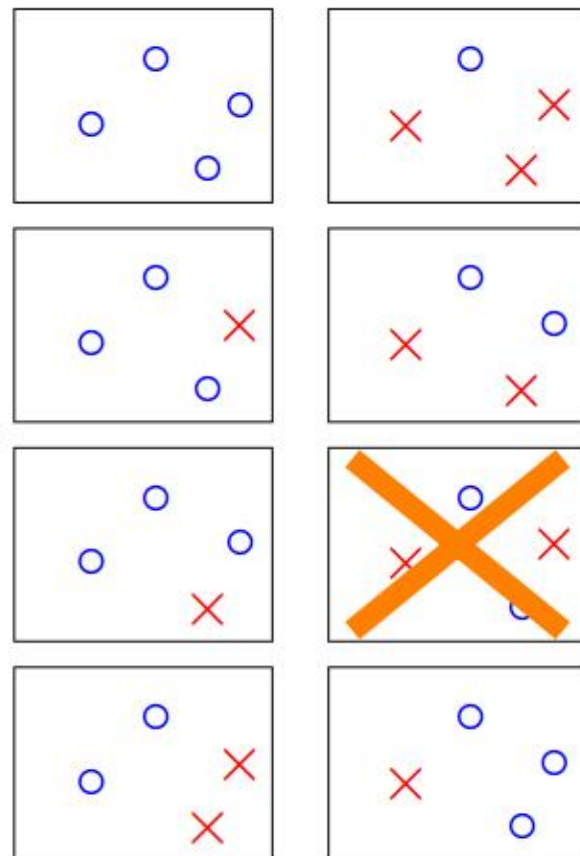
$$\mathcal{H} = \left\{ \text{all lines in } \mathbb{R}^2 \right\}$$

for four inputs $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$



for any four inputs
at most 14

14: $2 \times$



Fun Time

What is the effective number of lines for five inputs $\in \mathbb{R}^2$?

① 14

② 16

③ 22

④ 32

Fun Time

What is the effective number of lines for five inputs $\in \mathbb{R}^2$?

① 14

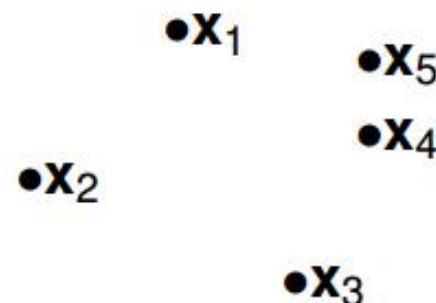
② 16

③ 22

④ 32

Reference Answer: ③

If you put the inputs roughly around a circle, you can then pick any consecutive inputs to be on one side of the line, and the other inputs to be on the other side. The procedure leads to effectively 22 kinds of lines, which is **much smaller than** $2^5 = 32$. You shall find it difficult to generate more kinds by varying the inputs, and we will give a formal proof in future lectures.



Dichotomies: Mini-hypotheses

$$\mathcal{H} = \{\text{hypothesis } h: \mathcal{X} \rightarrow \{\textcolor{red}{\times}, \textcolor{blue}{\circ}\}\}$$

- call

$$h(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = (h(\mathbf{x}_1), h(\mathbf{x}_2), \dots, h(\mathbf{x}_N)) \in \{\textcolor{red}{\times}, \textcolor{blue}{\circ}\}^N$$

a **dichotomy**: hypothesis 'limited' to the eyes of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$

- $\mathcal{H}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$:

all dichotomies 'implemented' by $\mathcal{H}$ on $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$

	hypotheses $\mathcal{H}$	dichotomies $\mathcal{H}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$
e.g.	all lines in $\mathbb{R}^2$	$\{\textcolor{blue}{\circ}\textcolor{blue}{\circ}\textcolor{blue}{\circ}\textcolor{blue}{\circ}, \textcolor{blue}{\circ}\textcolor{blue}{\circ}\textcolor{red}{\times}, \textcolor{blue}{\circ}\textcolor{red}{\times}\textcolor{red}{\times}, \dots\}$
size	possibly infinite	upper bounded by 2^N

$|\mathcal{H}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)|$: candidate for **replacing M**

Growth Function

- $|\mathcal{H}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)|$: depend on inputs $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$
- growth function:
remove dependence by **taking max of all possible** $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$

$$m_{\mathcal{H}}(N) = \max_{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N \in \mathcal{X}} |\mathcal{H}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)|$$

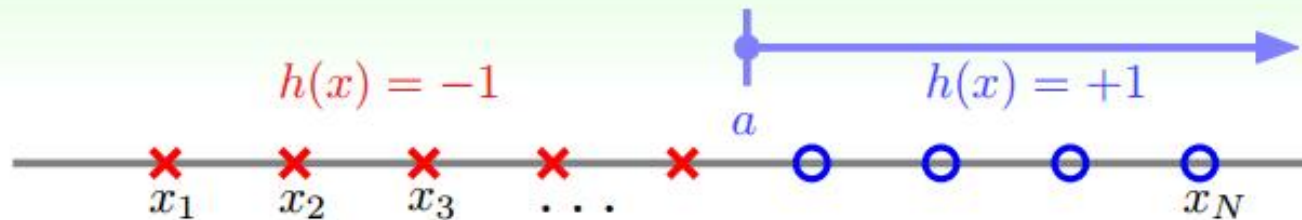
- finite, upper-bounded by 2^N

lines in 2D

N	$m_{\mathcal{H}}(N)$
1	2
2	4
3	$\max(\dots, 6, 8)$ $= 8$
4	$14 < 2^N$

how to 'calculate' the growth function?

Growth Function for Positive Rays



- $\mathcal{X} = \mathbb{R}$ (one dimensional)
- $\mathcal{H}$ contains h , where **each** $h(x) = \text{sign}(x - a)$ **for threshold** a
- 'positive half' of 1D perceptrons

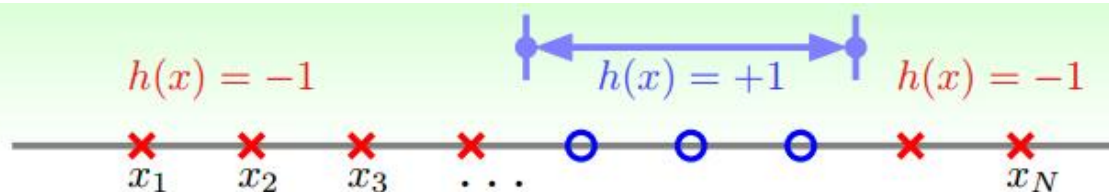
one dichotomy for $a \in$ each spot (x_n, x_{n+1}) :

$$m_{\mathcal{H}}(N) = N + 1$$

$(N + 1) \ll 2^N$ when N large!

x_1	x_2	x_3	x_4
o	o	o	o
x	o	o	o
x	x	o	o
x	x	x	o
x	x	x	x

Growth Function for Positive Intervals



- $\mathcal{X} = \mathbb{R}$ (one dimensional)
- $\mathcal{H}$ contains h , where **each** $h(x) = +1$ **iff** $x \in [\ell, r)$, **-1 otherwise**

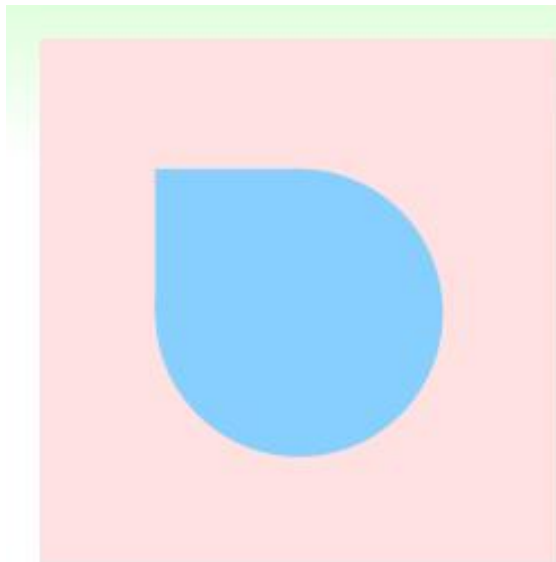
one dichotomy for each 'interval kind'

$$\begin{aligned}
 m_{\mathcal{H}}(N) &= \underbrace{\binom{N+1}{2}}_{\text{interval ends in } N+1 \text{ spots}} + \underbrace{1}_{\text{all } \times} \\
 &= \frac{1}{2}N^2 + \frac{1}{2}N + 1
 \end{aligned}$$

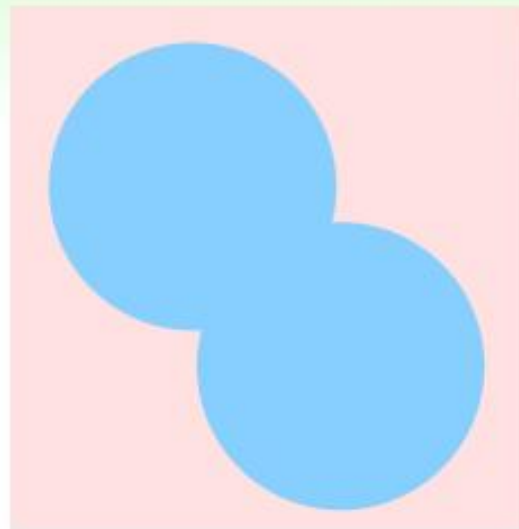
$$\left(\frac{1}{2}N^2 + \frac{1}{2}N + 1\right) \ll 2^N \text{ when } N \text{ large!}$$

x_1	x_2	x_3	x_4
o	x	x	x
o	o	x	x
o	o	o	x
o	o	o	o
x	o	x	x
x	o	o	x
x	o	o	o
x	x	o	x
x	x	o	o
x	x	x	o
x	x	x	x

Growth Function for Convex Sets(1/2)



convex region in blue



non-convex region

- $\mathcal{X} = \mathbb{R}^2$ (two dimensional)
- $\mathcal{H}$ contains h , where $h(\mathbf{x}) = +1$ iff $\mathbf{x}$ in a convex region, -1 otherwise

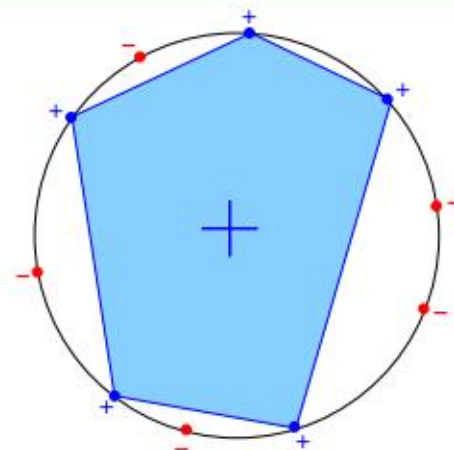
what is $m_{\mathcal{H}}(N)$?

Growth Function for Convex Sets(2/2)

- one possible set of N inputs:
 $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$ on a big circle
- **every dichotomy can be implemented**
by $\mathcal{H}$ using a convex region slightly
extended from **contour of positive inputs**

$$m_{\mathcal{H}}(N) = 2^N$$

- call those N inputs '**shattered**' by $\mathcal{H}$



$m_{\mathcal{H}}(N) = 2^N \iff$
exists N inputs that can be shattered

Fun Time

Consider positive **and negative** rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. The hypothesis set is often called '**decision stump**' to describe the shape of its hypotheses. What is the growth function $m_{\mathcal{H}}(N)$?

① N

② $N + 1$

③ $2N$

④ 2^N

Fun Time

Consider positive **and negative** rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. The hypothesis set is often called '**decision stump**' to describe the shape of its hypotheses. What is the growth function $m_{\mathcal{H}}(N)$?

① N

② $N + 1$

③ $2N$

④ 2^N

Reference Answer: ③

Two dichotomies when threshold in each of the $N - 1$ 'internal' spots; two dichotomies for the all-○ and all-× cases.

The Four Growth Functions

- positive rays: $m_{\mathcal{H}}(N) = N + 1$
- positive intervals: $m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1$
- convex sets: $m_{\mathcal{H}}(N) = 2^N$
- 2D perceptrons: $m_{\mathcal{H}}(N) < 2^N$ in some cases

what if $m_{\mathcal{H}}(N)$ replaces M ?

$$\mathbb{P} [|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon] \stackrel{?}{\leq} 2 \cdot m_{\mathcal{H}}(N) \cdot \exp(-2\epsilon^2 N)$$

polynomial: good; exponential: bad

for 2D or general perceptrons,
 $m_{\mathcal{H}}(N)$ **polynomial**?

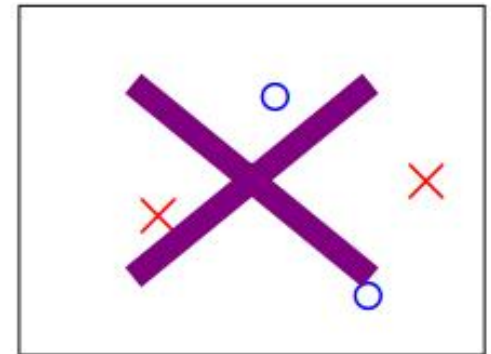
Break Point of H

what do we know about 2D perceptrons now?

three inputs: 'exists' shatter;
four inputs, 'for all' no shatter

if no k inputs can be shattered by $\mathcal{H}$,
call k a **break point** for $\mathcal{H}$

- $m_{\mathcal{H}}(k) < 2^k$
- $k + 1, k + 2, k + 3, \dots$ also break points!
- will study **minimum break point** k



2D perceptrons: **break point at 4**

The Four Break Points

- positive rays: $m_{\mathcal{H}}(N) = N + 1 = O(N)$
break point at 2
- positive intervals: $m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1 = O(N^2)$
break point at 3
- convex sets: $m_{\mathcal{H}}(N) = 2^N$
no break point
- 2D perceptrons: $m_{\mathcal{H}}(N) < 2^N$ in some cases
break point at 4

conjecture:

- no break point: $m_{\mathcal{H}}(N) = 2^N$ (sure!)
- break point k : $m_{\mathcal{H}}(N) = O(N^{k-1})$
excited? wait for next lecture :-)

The Four Break Points

growth function $m_{\mathcal{H}}(N)$: max number of dichotomies

- positive rays: $m_{\mathcal{H}}(N) = N + 1$
 $m_{\mathcal{H}}(2) = 3 < 2^2$: break point at 2
- positive intervals: $m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1$
 $m_{\mathcal{H}}(3) = 7 < 2^3$: break point at 3
- convex sets: $m_{\mathcal{H}}(N) = 2^N$ always: no break point

- 2D perceptrons: $m_{\mathcal{H}}(N) < 2^N$ in some cases
 $m_{\mathcal{H}}(4) = 14 < 2^4$: break point at 4

break point $k \implies$ break point $k + 1, \dots$
what else?

Fun Time

Consider positive **and negative** rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. As discussed in an earlier quiz question, the growth function $m_{\mathcal{H}}(N) = 2N$. What is the minimum break point for $\mathcal{H}$?

1 1

2 2

3 3

4 4

Fun Time

Consider positive **and negative** rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. As discussed in an earlier quiz question, the growth function $m_{\mathcal{H}}(N) = 2N$. What is the minimum break point for $\mathcal{H}$?

① 1

② 2

③ 3

④ 4

Reference Answer: ③

At $k = 3$, $m_{\mathcal{H}}(k) = 6$ while $2^k = 8$.

Bounding Function

bounding function $B(N, k)$:

maximum possible $m_{\mathcal{H}}(N)$ when break point = k

- combinatorial quantity:
maximum number of length- N vectors with $(\circ, \times)$
while ‘**no shatter**’ **any length- k** subvectors
- irrelevant of the details of $\mathcal{H}$
e.g. $B(N, 3)$ bounds both
 - positive intervals ($k = 3$)
 - 1D perceptrons ($k = 3$)

new goal: $B(N, k) \leq \text{poly}(N)$?

Table of Bounding Function

		k						
$B(N, k)$		1	2	3	4	5	6	...
N	1	1	2	2	2	2	2	...
	2	1	3	4	4	4	4	...
	3	1	4		8	8	8	...
	4	1				16	16	...
	5	1					32	...
	6	1						...
	$\vdots$	$\vdots$						

Known

- $B(N, k) = 2^N$ for $N < k$
—including all dichotomies not violating 'breaking condition'

Table of Bounding Function

$B(N, k)$		k						
		1	2	3	4	5	6	...
N	1	1	2	2	2	2	2	...
	2	1	3	4	4	4	4	...
	3	1	4	7	8	8	8	...
	4	1			15	16	16	...
	5	1				31	32	...
	6	1					63	...
	$\vdots$	$\vdots$						$\ddots$

Known

- $B(N, k) = 2^N - 1$ for $N = k$
—removing a single dichotomy satisfies 'breaking condition'

more than halfway done! :-)

Table of Bounding Function

$$\begin{aligned}
 B(4, 3) &= 2\alpha + \beta \\
 \alpha + \beta &\leq B(3, 3) \\
 \alpha &\leq B(3, 2) \\
 \Rightarrow B(4, 3) &\leq B(3, 3) + B(3, 2)
 \end{aligned}$$

		k					
$B(N, k)$		1	2	3	4	5	6
N	1	1	2	2	2	2	2
	2	1	3	4	4	4	4
	3	1	4	7	8	8	8
	4	1	≤ 5	11	15	16	16
	5	1	≤ 6	≤ 16	≤ 26	31	32
	6	1	≤ 7	≤ 22	≤ 42	≤ 57	63

now have **upper bound** of bounding function

Q?

For the 2D perceptrons, which of the following claim is true?

- ① minimum break point $k = 2$
- ② $m_{\mathcal{H}}(4) = 15$
- ③ $m_{\mathcal{H}}(N) < B(N, k)$ when $N = k =$ minimum break point
- ④ $m_{\mathcal{H}}(N) > B(N, k)$ when $N = k =$ minimum break point

Bounding Function: The Theorem

$$B(N, k) \leq \underbrace{\sum_{i=0}^{k-1} \binom{N}{i}}_{\text{highest term } N^{k-1}}$$

- simple induction using **boundary and inductive formula**
- for fixed k , $B(N, k)$ upper bounded by $\text{poly}(N)$
 $\implies m_{\mathcal{H}}(N)$ **is $\text{poly}(N)$ if break point exists**

' $\leq$ ' can be ' $=$ ' actually,
go play and prove it if math lover! :-)

The Three Break Points

$$B(N, k) \leq \underbrace{\sum_{i=0}^{k-1} \binom{N}{i}}_{\text{highest term } N^{k-1}}$$

- positive rays: $m_{\mathcal{H}}(N) = N + 1 \leq N + 1$
 $\circ \times \quad m_{\mathcal{H}}(2) = 3 < 2^2$: break point at 2
- positive intervals: $m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1 \leq \frac{1}{2}N^2 + \frac{1}{2}N + 1$
 $\circ \times \circ \quad m_{\mathcal{H}}(3) = 7 < 2^3$: break point at 3
- 2D perceptrons: $m_{\mathcal{H}}(N) = ? \leq \frac{1}{6}N^3 + \frac{5}{6}N + 1$
 $\times \begin{array}{c} \circ \\ \circ \end{array} \times \quad m_{\mathcal{H}}(4) = 14 < 2^4$: break point at 4

can bound $m_{\mathcal{H}}(N)$ by only **one break point**

BAD Bound for General H

want:

$$\mathbb{P}[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon] \leq 2 m_{\mathcal{H}}(N) \cdot \exp\left(-2 \epsilon^2 N\right)$$

actually, when N large enough,

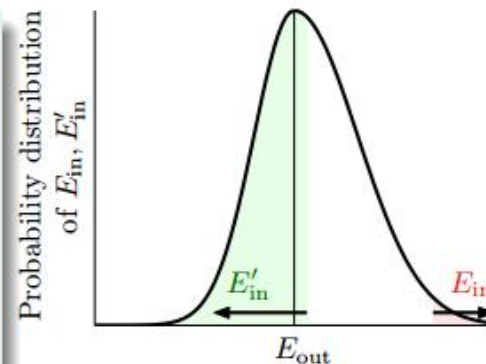
$$\mathbb{P}[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon] \leq 2 \cdot 2 m_{\mathcal{H}}(2N) \cdot \exp\left(-2 \cdot \frac{1}{16} \epsilon^2 N\right)$$

next: **sketch** of proof

Step 1: Replace E_{out} by E_{in}

$$\begin{aligned} & \frac{1}{2} \mathbb{P} \left[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon \right] \\ & \leq \mathbb{P} \left[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \frac{\epsilon}{2} \right] \end{aligned}$$

- $E_{\text{in}}(h)$ finitely many, $E_{\text{out}}(h)$ infinitely many
—replace the evil E_{out} first
- how? sample verification set $\mathcal{D}'$ of size N to calculate E'_{in}
- BAD h of $E_{\text{in}} - E_{\text{out}}$
probably $\Rightarrow$ BAD h of $E_{\text{in}} - E'_{\text{in}}$

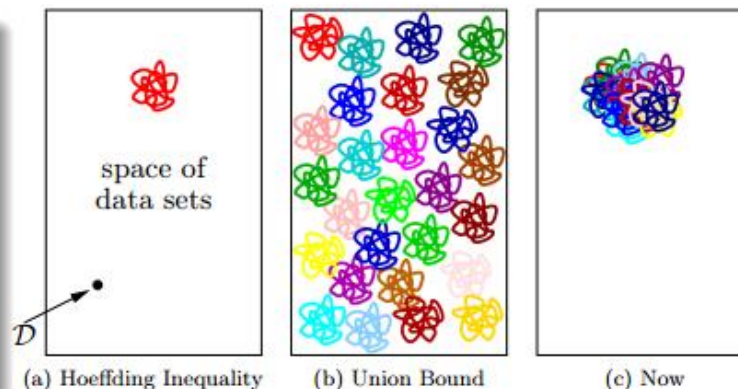


evil E_{out} removed by
verification with '**ghost data**'

Step 2: Decompose $\mathcal{H}$ by Kind

$$\begin{aligned}\text{BAD} &\leq 2\mathbb{P}\left[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \frac{\epsilon}{2}\right] \\ &\leq 2m_{\mathcal{H}}(2N)\mathbb{P}\left[\text{fixed } h \text{ s.t. } |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \frac{\epsilon}{2}\right]\end{aligned}$$

- E_{in} with $\mathcal{D}$, E'_{in} with $\mathcal{D}'$
—now $m_{\mathcal{H}}$ comes to play
- how? infinite $\mathcal{H}$ becomes $|\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{x}'_1, \dots, \mathbf{x}'_N)|$ kinds
- union bound on $m_{\mathcal{H}}(2N)$ kinds

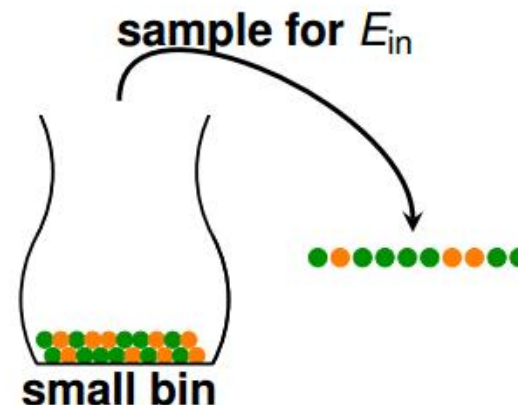


use $m_{\mathcal{H}}(2N)$ to calculate BAD-overlap properly

Step 3: Use Hoeffding without Replacement

$$\begin{aligned}\text{BAD} &\leq 2m_{\mathcal{H}}(2N)\mathbb{P}\left[\text{fixed } h \text{ s.t. } |E_{\text{in}}(h) - E'_{\text{in}}(h)| > \frac{\epsilon}{2}\right] \\ &\leq 2m_{\mathcal{H}}(2N) \cdot 2 \exp\left(-2\left(\frac{\epsilon}{4}\right)^2 N\right)\end{aligned}$$

- consider bin of $2N$ examples, choose N for E_{in} , leave others for E'_{in}
 $|E_{\text{in}} - E'_{\text{in}}| > \frac{\epsilon}{2} \Leftrightarrow \left|E_{\text{in}} - \frac{E_{\text{in}} + E'_{\text{in}}}{2}\right| > \frac{\epsilon}{4}$
- so? just 'smaller bin', 'smaller ϵ ', and Hoeffding without replacement



use Hoeffding after zooming to fixed h

Vapnik-Chervonenkis (VC) bound:

$$\begin{aligned} & \mathbb{P}\left[\exists h \in \mathcal{H} \text{ s.t. } |E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon\right] \\ & \leq 4m_{\mathcal{H}}(2N) \exp\left(-\frac{1}{8}\epsilon^2 N\right) \end{aligned}$$

- replace E_{out} by E'_{in}
- decompose $\mathcal{H}$ by kind
- use Hoeffding without replacement

2D perceptrons:

- break point? 4
- $m_{\mathcal{H}}(N)$? $O(N^3)$

learning with 2D perceptrons feasible! :-)

VC Dimension

the formal name of **maximum non-break** point

Definition

VC dimension of $\mathcal{H}$, denoted $d_{\text{vc}}(\mathcal{H})$ is

largest N for which $m_{\mathcal{H}}(N) = 2^N$

- the **most** inputs $\mathcal{H}$ that can shatter
- $d_{\text{vc}} = \text{'minimum } k' - 1$

$N \leq d_{\text{vc}} \implies \mathcal{H}$ can shatter some N inputs

$k > d_{\text{vc}} \implies k$ is a break point for $\mathcal{H}$

if $N \geq 2, d_{\text{vc}} \geq 2, m_{\mathcal{H}}(N) \leq N^{d_{\text{vc}}}$

The Four VC Dimensions

- positive rays:

$$d_{VC} = 1$$



$$m_{\mathcal{H}}(N) = N + 1$$

- positive intervals:

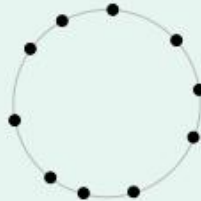
$$d_{VC} = 2$$



$$m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1$$

- convex sets:

$$d_{VC} = \infty$$



$$m_{\mathcal{H}}(N) = 2^N$$

- 2D perceptrons:

$$d_{VC} = 3$$



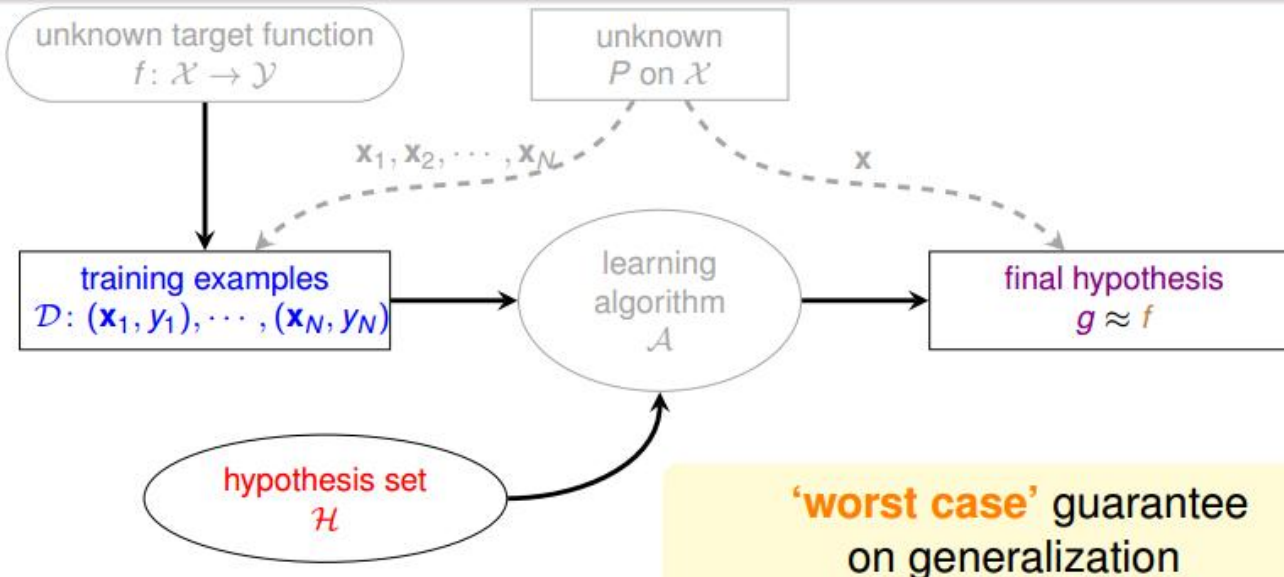
$$m_{\mathcal{H}}(N) \leq N^3 \text{ for } N \geq 2$$

good: **finite** d_{VC}

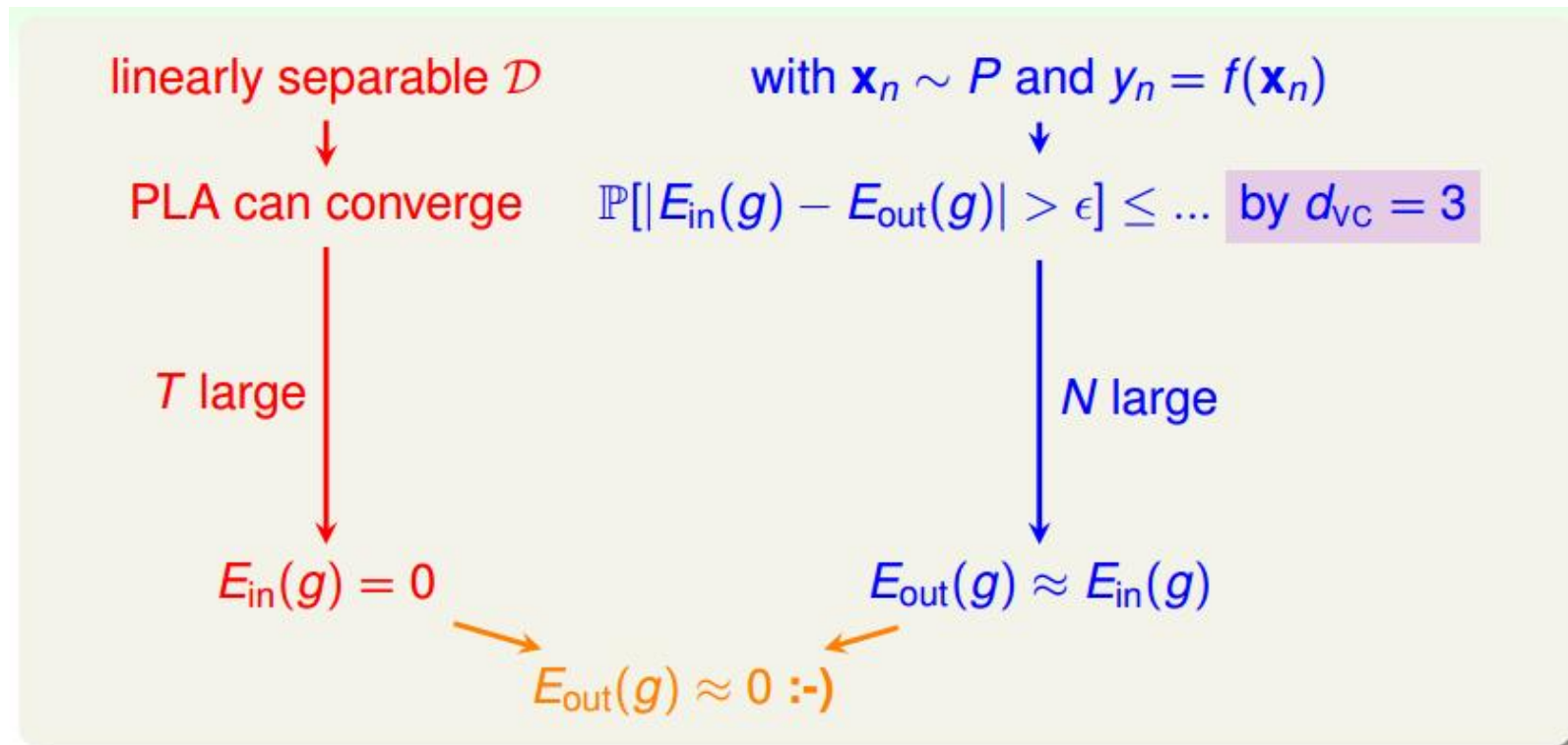
VC Dimension and Learning

finite $d_{VC} \implies g$ 'will' generalize ($E_{out}(g) \approx E_{in}(g)$)

- regardless of learning algorithm $\mathcal{A}$
- regardless of input distribution P
- regardless of target function f



2D PLA Revisited



general PLA for $\mathbf{x}$ with more than 2 features?

VC Dimension of Perceptrons

- 1D perceptron (pos/neg rays): $d_{VC} = 2$
- 2D perceptrons: $d_{VC} = 3$
 - $d_{VC} \geq 3$:

 - $d_{VC} \leq 3$:

- d -D perceptrons: $d_{VC} \stackrel{?}{=} d + 1$

two steps:

- $d_{VC} \geq d + 1$
- $d_{VC} \leq d + 1$

M and d_{VC}

- 1 can we make sure that $E_{\text{out}}(g)$ is close enough to $E_{\text{in}}(g)$?
- 2 can we make $E_{\text{in}}(g)$ small enough?

small M

- 1 Yes!,
 $\mathbb{P}[\mathbf{BAD}] \leq 2 \cdot M \cdot \exp(\dots)$
- 2 No!, too few choices

large M

- 1 No!,
 $\mathbb{P}[\mathbf{BAD}] \leq 2 \cdot M \cdot \exp(\dots)$
- 2 Yes!, many choices

small d_{VC}

- 1 Yes!, $\mathbb{P}[\mathbf{BAD}] \leq$
 $4 \cdot (2N)^{d_{VC}} \cdot \exp(\dots)$
- 2 No!, too limited power

large d_{VC}

- 1 No!, $\mathbb{P}[\mathbf{BAD}] \leq$
 $4 \cdot (2N)^{d_{VC}} \cdot \exp(\dots)$
- 2 Yes!, lots of power

using the right d_{VC} (or $\mathcal{H}$) is important

VC Bound Rephrase: Penalty for Model Complexity

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)| > \epsilon}_{\text{BAD}} \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

Rephrase

..., with probability $\geq 1 - \delta$, **GOOD**: $|E_{in}(g) - E_{out}(g)| \leq \epsilon$

$$\text{set } \delta = 4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\frac{\delta}{4(2N)^{d_{VC}}} = \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right) = \frac{1}{8}\epsilon^2 N$$

$$\sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)} = \epsilon$$

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For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)|}_{\text{BAD}} > \epsilon \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

Rephrase

..., with probability $\geq 1 - \delta$, **GOOD!**

$$\text{gen. error } |E_{in}(g) - E_{out}(g)| \leq \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)}$$

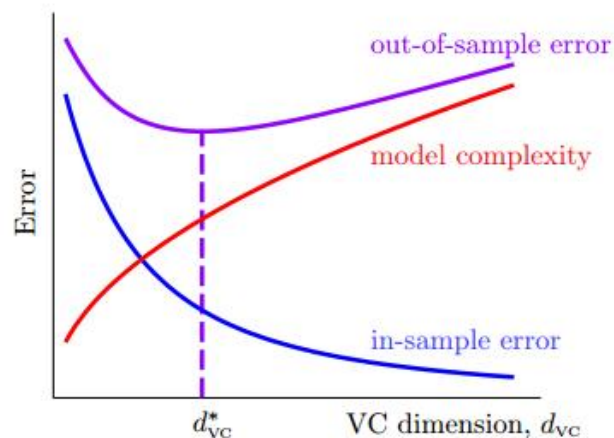
$$E_{in}(g) - \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)} \leq E_{out}(g) \leq E_{in}(g) + \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)}$$

$\underbrace{\sqrt{\dots}}_{\Omega(N, \mathcal{H}, \delta)}$: penalty for **model complexity**

THE VC Message

with a high probability,

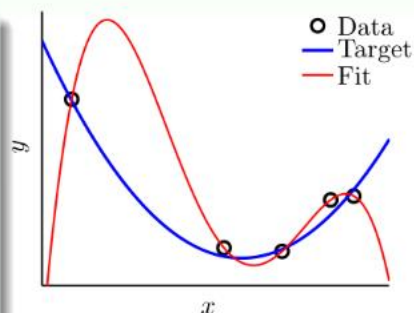
$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \underbrace{\sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{\text{VC}}}}{\delta} \right)}}_{\Omega(N, \mathcal{H}, \delta)}$$



- $d_{\text{VC}} \uparrow$: $E_{\text{in}} \downarrow$ but $\Omega \uparrow$
- $d_{\text{VC}} \downarrow$: $\Omega \downarrow$ but $E_{\text{in}} \uparrow$
- best d_{VC}^* in the middle

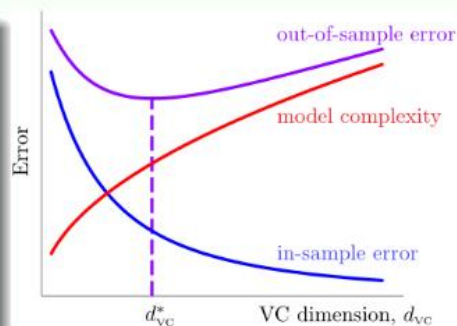
powerful $\mathcal{H}$ not always good!

- regression for $x \in \mathbb{R}$ with $N = 5$ examples
- target $f(x)$ = 2nd order polynomial
- label $y_n = f(x_n) + \text{very small noise}$
- linear regression in $\mathcal{Z}$ -space + Φ = 4th order polynomial
- unique solution passing all examples $\implies E_{\text{in}}(g) = 0$
- $E_{\text{out}}(g)$ huge



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- take $d_{\text{VC}} = 1126$ for learning:
bad generalization
— $(E_{\text{out}} - E_{\text{in}})$ large
- switch from $d_{\text{VC}} = d_{\text{VC}}^*$ to $d_{\text{VC}} = 1126$:
overfitting
— $E_{\text{in}} \downarrow, E_{\text{out}} \uparrow$
- switch from $d_{\text{VC}} = d_{\text{VC}}^*$ to $d_{\text{VC}} = 1$:
underfitting
— $E_{\text{in}} \uparrow, E_{\text{out}} \uparrow$



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VC Bound Rephrase: Sample Complexity

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2$, $d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)|}_{\text{BAD}} > \epsilon \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

given specs $\epsilon = 0.1$, $\delta = 0.1$, $d_{VC} = 3$, want $4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right) \leq \delta$

N	bound
100	2.82×10^7
1,000	9.17×10^9
10,000	1.19×10^8
100,000	1.65×10^{-38}
29,300	9.99×10^{-2}

sample complexity:
need $N \approx 10,000d_{VC}$ in theory

practical rule of thumb:

$$N \approx 10d_{VC} \text{ often enough!}$$

Looseness of VC Bound

$$\mathbb{P}_{\mathcal{D}} \left[|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon \right] \leq 4(2N)^{d_{\text{VC}}} \exp \left(-\frac{1}{8} \epsilon^2 N \right)$$

theory: $N \approx 10,000 d_{\text{VC}}$; practice: $N \approx 10 d_{\text{VC}}$

Why?

- Hoeffding for unknown E_{out} **any distribution, any target**
- $m_{\mathcal{H}}(N)$ instead of $|\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N)|$ **'any' data**
- $N^{d_{\text{VC}}}$ instead of $m_{\mathcal{H}}(N)$ **'any' $\mathcal{H}$ of same d_{VC}**
- union bound on worst cases **any choice made by $\mathcal{A}$**

— **but hardly better, and 'similarly loose for all models'**

philosophical message of VC bound
important for improving ML

Q?

Consider the VC Bound below. How can we decrease the probability of getting **BAD** data?

$$\mathbb{P}_{\mathcal{D}} \left[|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon \right] \leq 4(2N)^{d_{\text{vc}}} \exp \left(-\frac{1}{8} \epsilon^2 N \right)$$

- ① decrease model complexity d_{vc}
- ② increase data size N a lot
- ③ increase generalization error tolerance ϵ
- ④ all of the above



机器学习项目流程



机器学习一般步骤

数据搜集



数据清洗



特征工程



数据建模



机器学习一般步骤

